Model for "SrVO3"

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General Condition

- Basis type: `lg`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{1u}, A_{2g}, A_{2u}, E_g, E_u, T_{1g}, T_{1u}, T_{2g}, T_{2u}]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{1u}, A_{2g}, A_{2u}, E_g, E_u, T_{1g}, T_{1u}, T_{2g}, T_{2u}]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{1u}, A_{2g}, A_{2u}, E_g, E_u, T_{1g}, T_{1u}, T_{2g}, T_{2u}]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 221 O_{h}^{1} $Pm\bar{3}m$ [cubic]
- Associated point group: PG No. 221 O_{h} $m\bar{3}m$ [cubic]
- Unit cell:
  $a =$ `3.84000`, $b =$ `3.84000`, $c =$ `3.84000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `90.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 3.84000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[ 0.00000, 3.84000, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 3.84000]`

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1\|0\}$	2	$\{2_{001}\|0\}$	3	$\{2_{010}\|0\}$	4	$\{2_{100}\|0\}$	5	$\{3^{+}_{111}\|0\}$
6	$\{3^{+}_{-11-1}\|0\}$	7	$\{3^{+}_{1-1-1}\|0\}$	8	$\{3^{+}_{-1-11}\|0\}$	9	$\{3^{-}_{111}\|0\}$	10	$\{3^{-}_{1-1-1}\|0\}$
11	$\{3^{-}_{-1-11}\|0\}$	12	$\{3^{-}_{-11-1}\|0\}$	13	$\{2_{110}\|0\}$	14	$\{2_{1-10}\|0\}$	15	$\{4^{-}_{001}\|0\}$
16	$\{4^{+}_{001}\|0\}$	17	$\{4^{-}_{100}\|0\}$	18	$\{2_{011}\|0\}$	19	$\{2_{01-1}\|0\}$	20	$\{4^{+}_{100}\|0\}$
21	$\{4^{+}_{010}\|0\}$	22	$\{2_{101}\|0\}$	23	$\{4^{-}_{010}\|0\}$	24	$\{2_{-101}\|0\}$	25	$\{-1\|0\}$
26	$\{\mathrm{m}_{001}\|0\}$	27	$\{\mathrm{m}_{010}\|0\}$	28	$\{\mathrm{m}_{100}\|0\}$	29	$\{-3^{+}_{111}\|0\}$	30	$\{-3^{+}_{-11-1}\|0\}$
31	$\{-3^{+}_{1-1-1}\|0\}$	32	$\{-3^{+}_{-1-11}\|0\}$	33	$\{-3^{-}_{111}\|0\}$	34	$\{-3^{-}_{1-1-1}\|0\}$	35	$\{-3^{-}_{-1-11}\|0\}$
36	$\{-3^{-}_{-11-1}\|0\}$	37	$\{\mathrm{m}_{110}\|0\}$	38	$\{\mathrm{m}_{1-10}\|0\}$	39	$\{-4^{-}_{001}\|0\}$	40	$\{-4^{+}_{001}\|0\}$
41	$\{-4^{-}_{100}\|0\}$	42	$\{\mathrm{m}_{011}\|0\}$	43	$\{\mathrm{m}_{01-1}\|0\}$	44	$\{-4^{+}_{100}\|0\}$	45	$\{-4^{+}_{010}\|0\}$
46	$\{\mathrm{m}_{101}\|0\}$	47	$\{-4^{-}_{010}\|0\}$	48	$\{\mathrm{m}_{-101}\|0\}$				

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_{1g})$	A_{1g}	0	Q, T	-	-	1

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
2	$\mathbb{Q}_4(A_{1g})$	A_{1g}	4	Q, T	-	-	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$
3	$\mathbb{G}_3(A_{2g})$	A_{2g}	3	G, M	-	-	$\sqrt{15}xyz$
4	$\mathbb{Q}_3(A_{2u})$	A_{2u}	3	Q, T	-	-	$\sqrt{15}xyz$
5	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
6	$\mathbb{Q}_{2,2}(E_g)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
7	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\frac{-x^2-y^2+2z^2}{2}$
8	$\mathbb{G}_{2,2}(E_u)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
9	$\mathbb{G}_{1,1}(T_{1g})$	T_{1g}	1	G, M	-	1	x
10	$\mathbb{G}_{1,2}(T_{1g})$					2	y
11	$\mathbb{G}_{1,3}(T_{1g})$					3	z
12	$\mathbb{Q}_{1,1}(T_{1u})$	T_{1u}	1	Q, T	-	1	x
13	$\mathbb{Q}_{1,2}(T_{1u})$					2	y
14	$\mathbb{Q}_{1,3}(T_{1u})$					3	z
15	$\mathbb{Q}_{2,1}(T_{2g})$	T_{2g}	2	Q, T	-	1	$\sqrt{3}yz$
16	$\mathbb{Q}_{2,2}(T_{2g})$					2	$\sqrt{3}xz$
17	$\mathbb{Q}_{2,3}(T_{2g})$					3	$\sqrt{3}xy$
18	$\mathbb{G}_{2,1}(T_{2u})$	T_{2u}	2	G, M	-	1	$\sqrt{3}yz$
19	$\mathbb{G}_{2,2}(T_{2u})$					2	$\sqrt{3}xz$
20	$\mathbb{G}_{2,3}(T_{2u})$					3	$\sqrt{3}xy$
21	$\mathbb{Q}_{3,1}(T_{2u})$	T_{2u}	3	Q, T	-	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
22	$\mathbb{Q}_{3,2}(T_{2u})$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
23	$\mathbb{Q}_{3,3}(T_{2u})$					3	$\frac{\sqrt{15}z\left(x^2-y^2\right)}{2}$

Basis in full matrix

Table 3: dimension = 12

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	0	$\|p_x\rangle$ @O(1)	1	$\|p_y\rangle$ @O(1)	2	$\|p_z\rangle$ @O(1)	3	$\|p_x\rangle$ @O(2)	4	$\|p_y\rangle$ @O(2)
	5	$\|p_z\rangle$ @O(2)	6	$\|p_x\rangle$ @O(3)	7	$\|p_y\rangle$ @O(3)	8	$\|p_z\rangle$ @O(3)	9	$\|d_{xy}\rangle$ @V(1)
	10	$\|d_{xz}\rangle$ @V(1)	11	$\|d_{yz}\rangle$ @V(1)						

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z
$\|d_{xy}\rangle$	$\sqrt{3}xy$
$\|d_{xz}\rangle$	$\sqrt{3}xz$

continued ...

Table 4

orbital	definition
$\|d_{yz}\rangle$	$\sqrt{3}yz$

SAMB: 10 (all 10)

- `O` : 'O' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `3c`

  z1 $\quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

  z2 $\quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)\right)}{2}$

- `V` : 'V' site-cluster
  * bra: $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$
  * ket: $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$
  * wyckoff: `1b`

  z3 $\quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

- `O;O_001_1` : 'O'-'O' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `12b@12j`

  z4 $\quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

  z5 $\quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \dfrac{\sqrt{5}\left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g})\right)}{5}$

z6 $\mathbb{Q}_0^{(c)}(A_{1g}, c) = \dfrac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{1,1}^{(b)}(T_{1g}) + \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{1,2}^{(b)}(T_{1g}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{1,3}^{(b)}(T_{1g})\right)}{3}$

z7 $\mathbb{Q}_4^{(c)}(A_{1g}) = \dfrac{\sqrt{30}\left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g})\right)}{30}$

- `V;O_001_1` : 'O'-'V' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$
  * wyckoff: `6a@6f`

  z8 $\mathbb{Q}_0^{(c)}(A_{1g}) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{1,1}^{(b)}(T_{1u}) + \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{1,2}^{(b)}(T_{1u}) + \mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{1,3}^{(b)}(T_{1u})\right)}{3}$

- `V;V_001_1` : 'V'-'V' bond-cluster
  * bra: $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$
  * ket: $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$
  * wyckoff: `3a@3c`

  z9 $\mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

  z10 $\mathbb{Q}_0^{(c)}(A_{1g}, b) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)\right)}{2}$

Atomic SAMB

- bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
- ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$

x1 $\mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x2 $\mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$

x3 $\mathbb{Q}^{(a)}_{2,2}(E_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x4 $\mathbb{Q}^{(a)}_{2,1}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x5 $\mathbb{Q}^{(a)}_{2,2}(T_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x6 $\mathbb{Q}^{(a)}_{2,3}(T_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x7 $\mathbb{M}^{(a)}_{1,1}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x8 $\mathbb{M}^{(a)}_{1,2}(T_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x9 $\mathbb{M}^{(a)}_{1,3}(T_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$
- ket: $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$

x10 $\mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x11 $\mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} -\frac{\sqrt{6}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$

x12 $\mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$

x13 $\mathbb{Q}_{2,1}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x14 $\mathbb{Q}_{2,2}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x15 $\mathbb{Q}_{2,3}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x16 $\mathbb{M}_{1,1}^{(a)}(T_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x17 $\mathbb{M}_{1,2}^{(a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x18 $\mathbb{M}^{(a)}_{1,3}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

- bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
- ket: $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$

x19 $\mathbb{Q}^{(a)}_{3}(A_{2u}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$

x20 $\mathbb{Q}^{(a)}_{1,1}(T_{1u}) = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix}$

x21 $\mathbb{Q}^{(a)}_{1,2}(T_{1u}) = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

x22 $\mathbb{Q}^{(a)}_{1,3}(T_{1u}) = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$

x23 $\mathbb{Q}^{(a)}_{3,1}(T_{2u}) = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \end{bmatrix}$

x24 $\mathbb{Q}^{(a)}_{3,2}(T_{2u}) = \begin{bmatrix} -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

x25 $\mathbb{Q}^{(a)}_{3,3}(T_{2u}) = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$

x26 $\mathbb{G}^{(a)}_{2,1}(E_u) = \begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x27 $\mathbb{G}^{(a)}_{2,2}(E_u) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{3} & 0 & 0 \end{bmatrix}$

x28 $\mathbb{M}^{(a)}_{2,1}(E_u) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} \\ 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x29 $\mathbb{M}^{(a)}_{2,2}(E_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{3} & 0 & 0 \end{bmatrix}$

x30 $\mathbb{M}^{(a)}_{2,1}(T_{2u}) = \begin{bmatrix} 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 \\ 0 & -\frac{i}{2} & 0 \end{bmatrix}$

x31 $\mathbb{M}^{(a)}_{2,2}(T_{2u}) = \begin{bmatrix} -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{i}{2} \end{bmatrix}$

x32 $\mathbb{M}^{(a)}_{2,3}(T_{2u}) = \begin{bmatrix} 0 & \frac{i}{2} & 0 \\ 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 \end{bmatrix}$

x33 $\mathbb{T}_{3}^{(a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & \frac{\sqrt{6}i}{6} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 \end{bmatrix}$

x34 $\mathbb{T}_{1,1}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 \\ 0 & \frac{i}{2} & 0 \end{bmatrix}$

x35 $\mathbb{T}_{1,2}^{(a)}(T_{1u}) = \begin{bmatrix} \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{i}{2} \end{bmatrix}$

x36 $\mathbb{T}_{1,3}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & \frac{i}{2} & 0 \\ 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `1b`

y1 $\mathbb{Q}_{0}^{(s)}(A_{1g}) = [1]$

** Wyckoff: `3c`

y2 $\mathbb{Q}_{0}^{(s)}(A_{1g}) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y3 $\mathbb{Q}_{2,1}^{(s)}(E_{g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{3}\right]$

y4 $\mathbb{Q}_{2,2}^{(s)}(E_{g}) = \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0\right]$

- Bond cluster

** Wyckoff: `3a@3c`

y5 $\mathbb{Q}_{0}^{(b)}(A_{1g}) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y6 $\mathbb{Q}_{2,1}^{(b)}(E_{g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{3}\right]$

y7 $\mathbb{Q}_{2,2}^{(b)}(E_{g}) = \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0\right]$

y8 $\mathbb{T}_{1,1}^{(b)}(T_{1u}, a) = [i,\, 0,\, 0]$

y9 $\mathbb{T}_{1,2}^{(b)}(T_{1u}, a) = [0,\, i,\, 0]$

y10 $\mathbb{T}_{1,3}^{(b)}(T_{1u}, a) = [0,\, 0,\, i]$

** Wyckoff: `6a@6f`

y11 $\mathbb{Q}_{0}^{(b)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

y12 $\mathbb{T}_{0}^{(b)}(A_{1g}) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$

y13 $\mathbb{Q}_{2,1}^{(b)}(E_{g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right]$

y14 $\mathbb{Q}_{2,2}^{(b)}(E_{g}) = \left[-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0\right]$

y15 $\mathbb{T}_{2,1}^{(b)}(E_{g}, a) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{3}\right]$

y16 $\mathbb{T}_{2,2}^{(b)}(E_{g}, a) = \left[-\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, 0, 0\right]$

y17 $\mathbb{Q}^{(b)}_{1,1}(T_{1u}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0, 0, 0\right]$

y18 $\mathbb{Q}^{(b)}_{1,2}(T_{1u}) = \left[0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0\right]$

y19 $\mathbb{Q}^{(b)}_{1,3}(T_{1u}) = \left[0, 0, 0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

y20 $\mathbb{T}^{(b)}_{1,1}(T_{1u}, a) = \left[\frac{\sqrt{2}i}{2}, -\frac{\sqrt{2}i}{2}, 0, 0, 0, 0\right]$

y21 $\mathbb{T}^{(b)}_{1,2}(T_{1u}, a) = \left[0, 0, \frac{\sqrt{2}i}{2}, -\frac{\sqrt{2}i}{2}, 0, 0\right]$

y22 $\mathbb{T}^{(b)}_{1,3}(T_{1u}, a) = \left[0, 0, 0, 0, \frac{\sqrt{2}i}{2}, -\frac{\sqrt{2}i}{2}\right]$

** Wyckoff: `12b@12j`

y23 $\mathbb{Q}^{(b)}_{0}(A_{1g}) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

y24 $\mathbb{M}^{(b)}_{3}(A_{2g}) = \left[-\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}\right]$

y25 $\mathbb{Q}^{(b)}_{2,1}(E_g) = \left[\frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y26 $\mathbb{Q}^{(b)}_{2,2}(E_g) = \left[-\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, 0, 0, 0, 0\right]$

y27 $\mathbb{T}^{(b)}_{2,1}(E_g, a) = \left[\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, 0, 0, 0, 0\right]$

y28 $\mathbb{T}^{(b)}_{2,2}(E_g, a) = \left[\frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, \frac{\sqrt{6}i}{12}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

y29 $\mathbb{M}^{(b)}_{1,1}(T_{1g}) = \left[-\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, 0, 0, 0, 0, 0, 0, 0, 0\right]$

y30 $\mathbb{M}^{(b)}_{1,2}(T_{1g}) = \left[0, 0, 0, 0, -\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, 0, 0, 0, 0\right]$

y31 $\mathbb{M}^{(b)}_{1,3}(T_{1g}) = \left[0, 0, 0, 0, 0, 0, 0, 0, -\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

y32 $\mathbb{Q}^{(b)}_{1,1}(T_{1u}) = \left[0, 0, 0, 0, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}\right]$

y33 $\mathbb{Q}^{(b)}_{1,2}(T_{1u}) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, 0, 0, 0, 0, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}\right]$

y34 $\mathbb{Q}^{(b)}_{1,3}(T_{1u}) = \left[\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, 0, 0, 0, 0\right]$

y35 $\mathbb{T}^{(b)}_{1,1}(T_{1u}, a) = \left[0, 0, 0, 0, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}\right]$

y36 $\mathbb{T}^{(b)}_{1,2}(T_{1u}, a) = \left[-\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, 0, 0, 0, 0, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}\right]$

y37 $\mathbb{T}^{(b)}_{1,3}(T_{1u}, a) = \left[\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, 0, 0, 0, 0\right]$

y38 $\mathbb{Q}^{(b)}_{2,1}(T_{2g}) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0\right]$

y39 $\mathbb{Q}^{(b)}_{2,2}(T_{2g}) = \left[0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right]$

y40 $\mathbb{Q}^{(b)}_{2,3}(T_{2g}) = \left[0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y41 $\mathbb{M}^{(b)}_{2,1}(T_{2u}) = \left[0, 0, 0, 0, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}\right]$

y42 $\mathbb{M}^{(b)}_{2,2}(T_{2u}) = \left[-\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, 0, 0, 0, 0, -\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}\right]$

y43 $\mathbb{M}^{(b)}_{2,3}(T_{2u}) = \left[-\frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, -\frac{\sqrt{2}i}{4}, \frac{\sqrt{2}i}{4}, 0, 0, 0, 0\right]$

y44 $\mathbb{Q}^{(b)}_{3,1}(T_{2u}) = \left[0, 0, 0, 0, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}\right]$

y45 $\mathbb{Q}^{(b)}_{3,2}(T_{2u}) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, 0, 0, 0, 0, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}\right]$

y46 $\mathbb{Q}^{(b)}_{3,3}(T_{2u}) = \left[-\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, 0, 0, 0, 0\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	O	$\|p_x\rangle$, $\|p_y\rangle$, $\|p_z\rangle$
2	V	$\|d_{xy}\rangle$, $\|d_{xz}\rangle$, $\|d_{yz}\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	O	O	[1]	[p]	[p]

continued ...

Table 6

#	head	tail	neighbor	head (bra)	tail (ket)
2	O	V	[1]	[p]	[d]
3	V	V	[1]	[d]	[d]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'O' (#1) site cluster (3c), 4/mm.m

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.50000, 0.50000]	[1,2,3,4,17,18,19,20,25,26,27,28,41,42,43,44]
2	[0.50000, 0.00000, 0.50000]	[5,6,7,8,13,14,15,16,29,30,31,32,37,38,39,40]
3	[0.50000, 0.50000, 0.00000]	[9,10,11,12,21,22,23,24,33,34,35,36,45,46,47,48]

Table 8: 'V' (#2) site cluster (1b), m-3m

SL	position ($\boldsymbol{s}$)	mapping
1	[0.50000, 0.50000, 0.50000]	[1,2,3,4,$\cdots$,48]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'O'-'O' [1] (#1) bond cluster (12b@12j), ND, $|\boldsymbol{v}|$= 2.71529 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 0.50000,-0.50000]	[0.50000, 0.25000, 0.25000]	[1,-18,28,-43]	(3,1)	(2,1)	[0,0,0]
2	[0.00000,-0.50000,-0.50000]	[0.50000, 0.75000, 0.25000]	[2,-20,27,-41]	(3,1)	(2,1)	[0,1,0]
3	[0.00000, 0.50000, 0.50000]	[0.50000, 0.25000, 0.75000]	[3,-17,26,-44]	(3,1)	(2,1)	[0,0,-1]
4	[0.00000,-0.50000, 0.50000]	[0.50000, 0.75000, 0.75000]	[4,-19,25,-42]	(3,1)	(2,1)	[0,1,-1]
5	[-0.50000, 0.00000, 0.50000]	[0.25000, 0.50000, 0.25000]	[5,-15,32,-38]	(1,1)	(3,1)	[0,0,0]
6	[-0.50000, 0.00000,-0.50000]	[0.25000, 0.50000, 0.75000]	[6,-13,31,-40]	(1,1)	(3,1)	[0,0,1]
7	[0.50000, 0.00000, 0.50000]	[0.75000, 0.50000, 0.25000]	[7,-16,30,-37]	(1,1)	(3,1)	[-1,0,0]
8	[0.50000, 0.00000,-0.50000]	[0.75000, 0.50000, 0.75000]	[8,-14,29,-39]	(1,1)	(3,1)	[-1,0,1]
9	[0.50000,-0.50000, 0.00000]	[0.25000, 0.25000, 0.50000]	[9,-21,36,-48]	(2,1)	(1,1)	[0,0,0]
10	[-0.50000,-0.50000, 0.00000]	[0.75000, 0.25000, 0.50000]	[10,-23,35,-46]	(2,1)	(1,1)	[1,0,0]
11	[0.50000, 0.50000, 0.00000]	[0.25000, 0.75000, 0.50000]	[11,-22,34,-47]	(2,1)	(1,1)	[0,-1,0]
12	[-0.50000, 0.50000, 0.00000]	[0.75000, 0.75000, 0.50000]	[12,-24,33,-45]	(2,1)	(1,1)	[1,-1,0]

Table 10: 1-th 'O'-'V' [1] (#2) bond cluster (6a@6f), D, $|\boldsymbol{v}|$= 1.92 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.50000, 0.00000, 0.00000]	[0.25000, 0.50000, 0.50000]	[1,4,17,20,26,27,42,43]	(1,1)	(1,1)	[0,0,0]

continued ...

Table 10

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
2	[0.50000, 0.00000, 0.00000]	[0.75000, 0.50000, 0.50000]	[2,3,18,19,25,28,41,44]	(1,1)	(1,1)	[-1,0,0]
3	[0.00000,-0.50000, 0.00000]	[0.50000, 0.25000, 0.50000]	[5,8,13,16,30,31,38,39]	(2,1)	(1,1)	[0,0,0]
4	[0.00000, 0.50000, 0.00000]	[0.50000, 0.75000, 0.50000]	[6,7,14,15,29,32,37,40]	(2,1)	(1,1)	[0,-1,0]
5	[0.00000, 0.00000,-0.50000]	[0.50000, 0.50000, 0.25000]	[9,12,22,23,34,35,45,48]	(3,1)	(1,1)	[0,0,0]
6	[0.00000, 0.00000, 0.50000]	[0.50000, 0.50000, 0.75000]	[10,11,21,24,33,36,46,47]	(3,1)	(1,1)	[0,0,-1]

Table 11: 1-th 'V'-'V' [1] (#3) bond cluster (3a@3c), ND, $|\boldsymbol{v}|$= 3.84 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.00000, 0.50000, 0.50000]	[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]	(1,1)	(1,1)	[1,0,0]
2	[0.00000,-1.00000, 0.00000]	[0.50000, 0.00000, 0.50000]	[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]	(1,1)	(1,1)	[0,1,0]
3	[0.00000, 0.00000,-1.00000]	[0.50000, 0.50000, 0.00000]	[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]	(1,1)	(1,1)	[0,0,1]